STAT6611

- Lecture 16 (T, Oct 20)
- solutions posted to Homework 3
- Take home midterm due F, Oct 23 (midnight in Canvas)
- No new homework this week
  (next assignment next R, Oct 29)
  Homework 4

3.2 Moment Generating Function & Variance of a Quadratic Form

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is a $p \times 1$ vector. Take A to be a $p \times p$ symmetric matrix of constants. Then, for t such that $|t|$ is sufficiently small, it holds that

$$M_{\underline{x}' A \underline{x}}(t) = |I - 2tA\Sigma|^{-1/2} \exp\left[t\underline{\mu}' \left(I - 2tA\Sigma\right)^{-1} A\underline{\mu}\right].$$

Proof. For t with small sufficiently small $|t|$ and writing $\Sigma = \Gamma'\Gamma$ with $k \times p$ matrix Γ with $k = \text{rank}(\Sigma)$ and k-variate standard normal $\underline{z}$,

$$\begin{aligned}
M_{\underline{x}' A \underline{x}}(t) &= E \exp\left(t\underline{x}' A \underline{x}\right) = E \exp\left(t(\underline{\mu} + \Gamma'\underline{z})' A (\underline{\mu} + \Gamma'\underline{z})\right) \\
&= (2\pi)^{-k/2} \int_{\mathbb{R}^k} \exp\left[-\frac{1}{2}\underline{z}'(I - 2t\Gamma A\Gamma')\underline{z} + 2t\underline{z}'\Gamma A\underline{\mu} + t\underline{\mu}' A\underline{\mu}\right] d\underline{z} \\
&= |I - 2t\Gamma A\Gamma'|^{-1/2} \exp\left[2t^2\underline{\mu}' A\Gamma'(I - 2t\Gamma A\Gamma')^{-1}\Gamma A\underline{\mu} + t\underline{\mu}' A\underline{\mu}\right] \\
&= |I - 2tA\Sigma|^{-1/2} \exp\left[t\underline{\mu}'(I - 2tA\Sigma)^{-1} A\underline{\mu}\right]
\end{aligned}$$

applying Theorem 3.6/Lemma 3.1 for integration and re-arranging by corollary/Theorem 3.10 with $S = -2tA\Gamma'$, $U = \Gamma$, and by Theorem 1.5 with $R = I$, $S = -2tA\Gamma'$, $T = I$, and $V = \Gamma$, so $(I + SU)^{-1} = I - S(I + US)^{-1}U$. □

Based on this mgf of $\underline{x}' A \underline{x}$, the variance of $\underline{x}' A \underline{x}$ can be obtained, which is given by

$$\text{Var}(\underline{x}' A \underline{x}) = 2\text{tr}\left[(A\Sigma)^2\right] + 4\underline{\mu}' A\Sigma A\underline{\mu}.$$

Lemma 3.1. *If A is a $p \times p$ positive definite matrix and B is a $p \times p$ symmetric matrix, then there exists a $\delta > 0$ such that, for $|t| < \delta$, $A + tB$ is positive definite.*

Theorem 3.10. *Let R be a $p \times p$ matrix, T a $q \times q$ matrix, S a $p \times q$ matrix, and U a $q \times p$ matrix. If R and T are nonsingular, then*

$$|R + STU| = |R| \cdot |T| \cdot |T^{-1} + UR^{-1}S|.$$

Corollary. Let S be a $p \times q$ matrix and U a $q \times p$ matrix. Then, $|I + SU| = |I + US|$.

3.3 The Noncentral χ^2 and F Distributions

Definition. A random variable u is said to have a noncentral χ^2 distribution with degrees of freedom n and noncentrality parameter λ if u has the pdf

$$\chi^2(u\colon n,\lambda) = \begin{cases} \displaystyle\sum_{j=0}^{\infty} \left(\frac{e^{-\lambda}\lambda^j}{j!}\right) \left[\frac{u^{(n+2j-2)/2}e^{-u/2}}{\Gamma\left(\frac{n+2j}{2}\right)2^{j+(n/2)}}\right], & \text{for } u > 0, \\ 0, & \text{for } u \le 0, \end{cases}$$

where $\lambda \ge 0$ and $n > 0$.

Note. In the above pdf, $0! = 1$ and $0^0 = 1$.

With $\lambda = 0$, the noncentral χ^2 distribution reduces to the χ^2 or central χ^2 distribution with the pdf

$$\chi^2(u\colon n) = \begin{cases} \frac{u^{(n-2)/2}e^{-u/2}}{\Gamma\left(\frac{n}{2}\right)2^{n/2}}, & \text{for } u > 0, \\ 0 & \text{for } u \le 0. \end{cases}$$

Notation. We use $u \sim \chi^2(n,\lambda)$ to indicate that u is a random variable whose pdf is $\chi^2(u\colon n,\lambda)$, and $u \sim \chi^2(n)$ will be used if the pdf is $\chi^2(u\colon n)$.

Note that

$$\chi^2(u\colon n,\lambda) = \sum_{j=0}^{\infty} \left(\frac{e^{-\lambda}\lambda^j}{j!}\right) \chi^2(u\colon n+2j),$$

and that $e^{-\lambda}\lambda^j/j! = \Pr(v = j)$, where v is a Poisson random variable with parameter λ. Thus, $\chi^2(u\colon n,\lambda)$ is a pdf because $\chi^2(u\colon m)$ is a pdf for each $m > 0$.

Note: $\lambda=0$, $M_u(t)=(1-2t)^{-n/2}$, $t<\frac{1}{2}$ $\Rightarrow$ mgf of $\chi^2(n)$

For $u \sim \chi^2(n,\lambda)$, we have the mgf of u as

$$M_u(t) = (1-2t)^{-n/2} \exp\left(\frac{2t\lambda}{1-2t}\right), \quad t < \frac{1}{2}.$$

$$E e^{tu} = \int_0^\infty e^{tu} \sum_{j=0}^\infty \frac{e^{-\lambda}\lambda^j}{j!} \chi^2(u: n+2j)\, du$$

$$= \sum_{j=0}^\infty \frac{e^{-\lambda}\lambda^j}{j!} \int_0^\infty e^{tu} \chi^2(u: n+2j)\, du = \sum_{j=0}^\infty \frac{e^{-\lambda}\lambda^j}{j!} (1-2t)^{-\frac{n}{2}} (1-2t)^{-j}$$

$\int_0^\infty e^{tu}\chi^2(u:n+2j)\,du$: mgf $(1-2t)^{-\frac{(n+2j)}{2}}$ of $\chi^2(n+2j)$

use next: $\sum_{j=1}^\infty \frac{\lambda^j}{j!}(1-2t)^{-j} = \exp\left(\frac{\lambda}{1-2t}\right)$

Theorem 3.11. *Let $\underline{x} \sim N(\underline{\mu}, I_n)$. Define w by $w = \underline{x}'\underline{x}$. Then, $w \sim \chi^2(n, \frac{1}{2}\underline{\mu}'\underline{\mu})$.*

($E\underline{x}=\underline{\mu}$ may not be 0; length of $\underline{x}$; $\frac{1}{2}\underline{\mu}'\underline{\mu}$ = λ)

Proof. Let $u \sim \chi^2(n, \frac{1}{2}\underline{\mu}'\underline{\mu})$. By Theorem 3.1, it suffices to show that the mgf of w is the same as that of u. From the mgf of a quadratic form in Section 3.2, with $A = I_n$ and $\Sigma = I_n$, we have, for $t < 1/2$,

($\underline{x}'A\underline{x} = \underline{x}'\underline{x}$, $A=I_n$, $\Sigma=I_n$; result on mgf of quad. form)

$$M_w(t) = M_{\underline{x}'\underline{x}}(t) = |I_n - 2tI_n|^{-1/2} \exp\left[t\underline{\mu}'(I_n - 2tI_n)^{-1}\underline{\mu}\right]$$

($|I_n - 2tI_n| = |(1-2t)I_n| = (1-2t)^n$; $(I_n-2tI_n)^{-1} = (1-2t)^{-1}I_n = \frac{1}{1-2t}I_n$)

$$= (1-2t)^{-n/2} \exp\left[\frac{2t\left(\frac{1}{2}\underline{\mu}'\underline{\mu}\right)}{1-2t}\right] = M_u(t).$$

(mgf of $\chi^2(n, \lambda = \frac{1}{2}\underline{\mu}'\underline{\mu})$)

□

(Since we can find mean/variance of quad form $\underline{x}'A\underline{x}$ for normal $\underline{x} \sim N(\underline{\mu}, \Sigma)$, we can figure out mean/variance of $\chi^2(n,\lambda)$)

Let $u \sim \chi^2(n,\lambda)$ and $\underline{x} \sim N\left([2\lambda/n]^{1/2}\underline{1}_n, I_n\right)$, where $\underline{1}_n$ is the $n \times 1$ vector of 1's. Then,

($\underline{x}$; $\underline{\mu}$)

$$\frac{1}{2}\underline{\mu}'\underline{\mu} = \frac{1}{2}\left([2\lambda/n]^{1/2}\underline{1}_n\right)'\left([2\lambda/n]^{1/2}\underline{1}_n\right) = (\lambda/n)\,\underline{1}_n'\underline{1}_n = \lambda.$$

By Theorem 3.11, $\underline{x}'\underline{x} \sim \chi^2(n,\lambda)$ and so $\underline{x}'\underline{x} \stackrel{d}{=} u$. We then have moments for u.

(use this to get mean/variance of $u \sim \chi^2(n,\lambda)$)

By Lemma 2.8, we have $E\left(\underline{x}'A\underline{x}\right) = \mathrm{tr}\left(A\Sigma\right) + \underline{\mu}'A\underline{\mu}$. For so $u \sim \chi^2(n,\lambda)$,

$$E(u) = E(\underline{x}'\underline{x}) = \mathrm{tr}\left(I_n\right) + \left(2\lambda/n\right)\underline{1}_n'\underline{1}_n = n + 2\lambda.$$

By the result for $\mathrm{Var}(\underline{x}'A\underline{x})$ in Section 3.2,

$$\mathrm{Var}(u) = \mathrm{Var}(\underline{x}'\underline{x}) = 2\mathrm{tr}(I_n) + 4\left(2\lambda/n\right)\underline{1}_n'\underline{1}_n = 2(n+4\lambda).$$

Theorem 3.12. *If $u_1, \ldots, u_m$ are mutually independent random variables and if $u_i \sim \chi^2(n_i, \lambda_i)$ for $i = 1, \ldots, m$, then $u = u_1 + \cdots + u_m \sim \chi^2(n, \lambda)$, where $n = n_1 + \cdots + n_m$ and $\lambda = \lambda_1 + \cdots + \lambda_m$.*

Proof. Let $v \sim \chi^2(n,\lambda)$. By Theorem 3.1, it suffices to show that the mgf of u is the same as that of v. We have

$$M_u(t) = E\left\{\exp\left[t(u_1 + \cdots + u_m)\right]\right\} = \prod_{i=1}^{m} E\left[\exp\left(tu_i\right)\right] = \prod_{i=1}^{m}\left[\left(1-2t\right)^{-n_i/2}\exp\left(\frac{2t\lambda_i}{1-2t}\right)\right]$$

$$= \left(1-2t\right)^{-\left(\sum_{i=1}^m n_i\right)/2}\exp\left(\frac{2t\sum_{i=1}^m \lambda_i}{1-2t}\right) = \left(1-2t\right)^{-n/2}\exp\left(\frac{2t\lambda}{1-2t}\right) = M_v(t).$$

□

Theorem 3.13. *Let* $\underline{x} \sim N(\underline{\mu}, \Sigma)$, *where* $\underline{x}$ *is an* $n \times 1$ *vector and* $\text{rank}(\Sigma) = n$. *Define* w *by* $w = \underline{x}'\Sigma^{-1}\underline{x}$. *Then,* $w \sim \chi^2(n, \frac{1}{2}\underline{\mu}'\Sigma^{-1}\underline{\mu})$.

note: not necessary to be $\Sigma = I_n$

i.e., Σ is invertible or full rank

← this extends prior result when $\Sigma = I_n$ (the results match)

proof: Write $\Sigma = \Gamma'\Gamma$ where Γ' is $n \times n$ of rank n too.

Then $\underline{z} = (\Gamma')^{-1}\underline{x} \sim$ MVN too (since $\underline{x}$ is MVN)

with $E(\underline{z}) = E((\Gamma')^{-1}\underline{x}) = (\Gamma')^{-1}\underline{\mu} = \underline{\tilde{\mu}}$

& $\text{Var}(\underline{z}) = \text{Var}((\Gamma')^{-1}\underline{x})$

$= (\Gamma')^{-1}\text{Var}(\underline{x})\Gamma^{-1} = (\Gamma')^{-1}\Gamma'\Gamma\Gamma^{-1} = I_n$

$\Sigma = \Gamma'\Gamma$ using $(\Gamma')^{-1} = (\Gamma^{-1})'$ above or $((\Gamma')^{-1})' = \Gamma^{-1}$

So, $\underline{x}'\Sigma^{-1}\underline{x} = \underline{x}'\Gamma^{-1}(\Gamma')^{-1}\underline{x} = \underline{z}'\underline{z} \sim \chi^2(n, \frac{1}{2}\underline{\tilde{\mu}}'\underline{\tilde{\mu}})$ $\underline{\mu}\Sigma^{-1}\underline{\mu}$

by theorem 3.12 since $\underline{z} \sim \text{MVN}(\underline{\tilde{\mu}}, I_n)$

Definition. Let u_1 and u_2 be independent random variables such that $u_1 \sim \chi^2(n_1, \lambda)$ and $u_2 \sim \chi^2(n_2)$, where $\lambda \geq 0$, $n_1 > 0$, and $n_2 > 0$. Let

↑ non-central $\chi^2_{n_1}$ ↑ central $\chi^2_{n_2}$

$$w = \frac{u_1/n_1}{u_2/n_2}.$$

The distribution of w is called a noncentral F distribution with degrees of freedom n_1 and n_2 and noncentrality parameter λ. With $\lambda = 0$, the noncentral F distribution reduces to the F or central F distribution.

An mgf does not exist for the noncentral F distribution (including the central F distribution).

3.4 Distributions of Quadratic Forms

Note: since $\underline{x}'A\underline{x} \in \mathbb{R}$, so $\underline{x}'A\underline{x} = (\underline{x}'A\underline{x})' = \underline{x}'A'\underline{x}$

For any quadratic form $\underline{x}'A\underline{x}$, we have

$$\underline{x}'A\underline{x} = \left[\underline{x}'A\underline{x} + \left(\underline{x}'A\underline{x}\right)'\right]/2 = \left(\underline{x}'A\underline{x} + \underline{x}'A'\underline{x}\right)/2 = \underline{x}'\left[(A+A')/2\right]\underline{x} = \underline{x}'A_1\underline{x},$$

(so take average; same)

where $A_1 = (A+A')/2$ is symmetric. Thus, without loss of generality, in our discussion of distributions of quadratic forms, **we assume that the matrix A associated with any quadratic form $\underline{x}'A\underline{x}$ is symmetric.**

if A is not symmetric, replace A with symmetric $A_1 = (A+A')/2$ (doesn't change $\underline{x}'A_1\underline{x} = \underline{x}'A\underline{x}$)

Theorem 3.14. *If $\underline{x} \sim N(\underline{\mu}, \Sigma)$, then a sufficient condition for a quadratic form $\underline{x}'A\underline{x}$ and a vector $B\underline{x}$ of linear forms to be independent is that $B\Sigma A = 0$.*

(quad form; linear form; Note: $A' = A$)

proof: since $\underline{x} \sim MVN$, then so is $\begin{pmatrix} A\underline{x} \\ B\underline{x} \end{pmatrix} \sim MVN$

If $B\Sigma A = 0$, then $\text{cov}(B\underline{x}, A\underline{x}) = B\Sigma A' = B\Sigma A = 0$ (↑ $\text{Var}(\underline{x})$)

$\Rightarrow$ $A\underline{x}$ & $B\underline{x}$ are independent (by MVN + zero covariance)

$\Rightarrow$ $(A\underline{x})'A^-(A\underline{x})$ & $B\underline{x}$ are independent too (↑ function of $A\underline{x}$)

& $(A\underline{x})'A^-(A\underline{x}) = \underline{x}'A'A^-A\underline{x} = \underline{x}'AA^-A\underline{x} = \underline{x}'A\underline{x}$

Corollary. If $\underline{x} \sim N(\underline{\mu}, \Sigma)$, then a sufficient condition for the quadratic forms $\underline{x}'A\underline{x}$ and $\underline{x}'B\underline{x}$ to be independent is that $B\Sigma A = 0$. $= A\Sigma B$

(quad form; quad form)

Note: here both $A' = A$ & $B' = B$

so $B\Sigma A = 0 \Leftrightarrow (B\Sigma A)' = A\Sigma B = 0$

proof: $B\Sigma A = 0$ holds

then $A\underline{x}$ & $B\underline{x}$ are independent (as above)

$\Rightarrow$ $(A\underline{x})'A^-(A\underline{x})$ & $(B\underline{x})'B^-(B\underline{x})$ are independent too

($\underline{x}'A\underline{x}$; $\underline{x}'B\underline{x}$)

Above if $\underline{x} \sim N(\underline{\mu}, \Sigma)$ & Σ is invertible, then $\underline{x}'A\underline{x}$ & $\underline{x}'B\underline{x}$ are independent $\Leftrightarrow B\Sigma A = 0$

Remarks. If Σ is nonsingular, it can be shown that the condition $B\Sigma A = 0$ in Theorem 3.14 and the corollary to Theorem 3.15 is necessary as well as sufficient. The proof of necessity is much more difficult than that of sufficiency, see the paper by Harville and Kempthorne, "An alternative way to establish the necessity part of the classical result on the statistical independence of quadratic forms" (*Linear Algebra and Its Applications*, Volume 264, 1997, 205–215). If Σ is singular, the condition $B\Sigma A = 0$ in Theorem 3.14 and the corollary to Theorem 3.15 is no longer necessary.

← a generalization of past corollary

Theorem 3.15. *If $\underline{x} \sim N(\underline{\mu}, \Sigma)$, then a sufficient condition for the quadratic forms $\underline{x}'A_1\underline{x}$, ..., $\underline{x}'A_k\underline{x}$ to be mutually independent is that $A_i\Sigma A_j = 0$ for $i = 1, \ldots, k$, $j = 1, \ldots, k$, and $i \neq j$.*

↑ ↑ each pair of A_i & A_j $i \neq j$

Proof. By Theorem 3.5,

Same proof as before

$$\begin{bmatrix} A_1\underline{x} \\ \vdots \\ A_k\underline{x} \end{bmatrix} = \begin{bmatrix} A_1 \\ \vdots \\ A_k \end{bmatrix} \underline{x} \sim \text{MVN}$$

has a multivariate normal distribution. If $A_i\Sigma A_j = 0$ for $i = 1, \ldots, k$, $j = 1, \ldots, k$, and $i \neq j$, then $\text{cov}(A_i\underline{x}, A_j\underline{x}) = A_i\Sigma A_j' = A_i\Sigma A_j = 0$ for all $i \neq j$. Then, by Theorem 3.7, $A_1\underline{x}, \ldots, A_k\underline{x}$ are mutually independent. This implies that $\underline{x}'A_1\underline{x}, \ldots, \underline{x}'A_k\underline{x}$ are mutually independent, because for $i = 1, \ldots, k$,

$$\underline{x}'A_i\underline{x} = \underline{x}'A_iA_i^-A_i\underline{x} = (A_i'\underline{x})'A_i^-(A_i\underline{x}) = (A_i\underline{x})'A_i^-(A_i\underline{x})$$

is a function of $A_i\underline{x}$. □